Homework #3

3.3. Verify that (3-35) and (3-36) are solutions to (3-34).

$$A_{z1} = C_1 \frac{e^{-jkr}}{r} \tag{3-35}$$

$$A_{z2} = C_2 \frac{e^{+jkr}}{r} \tag{3-36}$$

$$\frac{d^2A_z(r)}{dr^2} + \frac{2}{r}\frac{dA_z(r)}{dr} + k^2A_z(r) = 0 \tag{3-34}$$

解: $A_{z1} = C_1 \frac{e^{-jkr}}{r}$ 时

$$\frac{d^2A_{z1}}{dr^2} = C_1 \frac{d}{dr}\left[-jk\frac{e^{-jkr}}{r} - \frac{e^{-jkr}}{r^2}\right]$$

$$= C_1\left[(-jk)^2\frac{e^{-jkr}}{r} + jk\frac{e^{-jkr}}{r^2} + jk\frac{e^{-jkr}}{r^2} + 2\frac{e^{-jkr}}{r^3}\right]$$

$$= C_1\left[-k^2\frac{e^{-jkr}}{r} + 2jk\frac{e^{-jkr}}{r^2} + \frac{2e^{-jkr}}{r^3}\right]$$

$$\frac{2}{r}\frac{dA_{z1}(r)}{dr} = \frac{2}{r}C_1\left(-jk\frac{e^{-jkr}}{r} - \frac{e^{-jkr}}{r^2}\right)$$

$$= C_1\left(-2jk\frac{e^{-jkr}}{r^2} - 2\frac{e^{-jkr}}{r^3}\right)$$

$$k^2A_{z1}(r) = C_1\left(k^2\frac{e^{-jkr}}{r}\right)$$

故 $A_{z1} = C_1\frac{e^{-jkr}}{r}$ 时, $\frac{d^2A_{z1}(r)}{dr^2} + \frac{2}{r}\frac{dA_{z1}(r)}{dr} + k^2A_{z1}(r) = 0$

$A_{z2} = C_2\frac{e^{jkr}}{r}$ 时

$$\frac{d^2A_{z2}}{dr^2} = C_2\frac{d}{dr}\left[jk\frac{e^{jkr}}{r} - \frac{e^{jkr}}{r^2}\right]$$

$$= C_2\left[(jk)^2\frac{e^{jkr}}{r} - jk\frac{e^{jkr}}{r^2} - jk\frac{e^{jkr}}{r^2} + \frac{2e^{jkr}}{r^3}\right]$$

$$= C_2\left[-k^2\frac{e^{jkr}}{r} - 2jk\frac{e^{jkr}}{r^2} + 2\frac{e^{jkr}}{r^3}\right]$$

$$\frac{2}{r}\frac{dA_{z2}}{dr} = C_2\left[2jk\frac{e^{jkr}}{r^2} - 2\frac{e^{jkr}}{r^3}\right]$$

$$k^2A_{z2}(r) = C_2k^2\frac{e^{jkr}}{r}$$

故 $A_{z2} = C_2\frac{e^{jkr}}{r}$ 时, $\frac{d^2A_{z2}(r)}{dr^2} + \frac{2}{r}\frac{dA_{z2}(r)}{dr} + k^2A_{z2}(r) = 0$

3.6. Derive (3-60) and (3-61).

$$-\nabla \cdot (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) = \mathbf{E}_1 \cdot \mathbf{J}_2 + \mathbf{H}_2 \cdot \mathbf{M}_1 - \mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{H}_1 \cdot \mathbf{M}_2 \quad (3\text{-}60)$$

$$-\oint\!\!\!\!\!\!\diagup_S (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) \cdot d\mathbf{s}'$$

$$= \iiint_V (\mathbf{E}_1 \cdot \mathbf{J}_2 + \mathbf{H}_2 \cdot \mathbf{M}_1 - \mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{H}_1 \cdot \mathbf{M}_2)\, dv' \quad (3\text{-}61)$$

解：令 $(J_1, M_1) \to (E_1, H_1)$，$(J_2, M_2) \to (E_2, H_2)$

则 $\begin{cases} \nabla \times E_1 = -j\omega\mu H_1 - M_1 \\ \nabla \times H_1 = j\omega\epsilon E_1 + J_1 \end{cases}$ $\begin{cases} \nabla \times E_2 = -j\omega\mu H_2 - M_2 \\ \nabla \times H_2 = j\omega\epsilon E_2 + J_2 \end{cases}$

由 $\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$

令 $A = E_1$，$B = H_2$

$$\nabla \cdot (E_1 \times H_2) = H_2 \cdot (\nabla \times E_1) - E_1 \cdot (\nabla \times H_2)$$

$$= H_2 \cdot (-j\omega\mu H_1 - M_1) - E_1 \cdot (j\omega\epsilon E_2 + J_2)$$

$$= -j\omega\mu H_1 \cdot H_2 - H_2 \cdot M_1 - j\omega\epsilon E_1 \cdot E_2 - E_1 \cdot J_2 \quad ①$$

令 $A = E_2$，$B = H_1$

同理 $\nabla \cdot (E_2 \times H_1) = -j\omega\mu H_1 \cdot H_2 - H_1 \cdot M_2 - j\omega\epsilon E_1 \cdot E_2 - E_2 \cdot J_1$ ②

-①+②，得 $-\nabla \cdot (E_1 \times H_2 - E_2 \times H_1) = H_2 \cdot M_1 - H_1 \cdot M_2 + E_1 \cdot J_2 - E_2 \cdot J_1$

整理，得 $-\nabla \cdot (E_1 \times H_2 - E_2 \times H_1) = E_1 \cdot J_2 + H_2 \cdot M_1 - E_2 \cdot J_1 - H_1 \cdot M_2$，证毕

由 $\iiint_V (\nabla \cdot F)\, dv' = \oint_S (F \cdot ds')$，

$$\iiint_V [-\nabla \cdot (E_1 \times H_2 - E_2 \times H_1)]\, dv' = -\oint_S (E_1 \times H_2 - E_2 \times H_1) \cdot ds'$$

对式(3.60)两边求积分，代入上式，得

$$-\oint_S (E_1 \times H_2 - E_2 \times H_1)\, ds' = \iiint_V (E_1 \cdot J_2 + H_2 \cdot M_1 - E_2 \cdot J_1 - H_1 \cdot M_2)\, dv'$$ 证毕

第三版第四章问题：4.1, 4.5 (only calculate (a)), 4.29，或第四版第四章问题：4.1, 4.5 (only calculate (a)), 4.31

4.1. A horizontal infinitesimal electric dipole of constant current I_0 is placed symmetrically about the origin and directed along the x-axis. Derive the

(a) far-zone fields radiated by the dipole

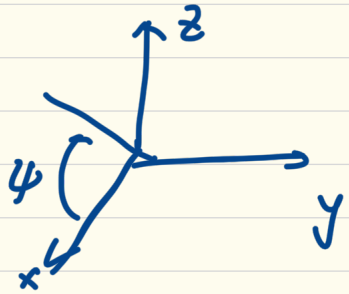
(b) directivity of the antenna

解：1(a) $\sin\psi = \sqrt{1-\cos^2\psi} = \sqrt{1-(\sin\theta\cdot\cos\phi)^2}$

故 $E_\psi = j\eta \frac{kIl_0 e^{-jkr}}{4\pi r}\cdot\sin\psi$

$= j\eta \frac{kIl_0 e^{-jkr}}{4\pi r}\sqrt{1-(\sin\theta\cos\phi)^2}$

$H_{x0} = \frac{E_\psi}{\eta} = j\frac{kIl_0 e^{-jkr}}{4\pi r}\sqrt{1-(\sin\theta\cos\phi)^2}$



(b) $U = U_0(1-\sin^2\theta\cos^2\phi)$

$P_{rad} = U_0\int_0^{2\pi}\int_0^{\pi}(1-\sin^2\theta\cdot\cos^2\phi)\cdot\sin\theta\,d\theta\,d\phi = \frac{8}{3}\pi U_0$

故 $D_0 = \frac{4\pi U_0}{P_{rad}} = \frac{4\pi U_0}{\frac{8}{3}\pi U_0} = \frac{3}{2}$

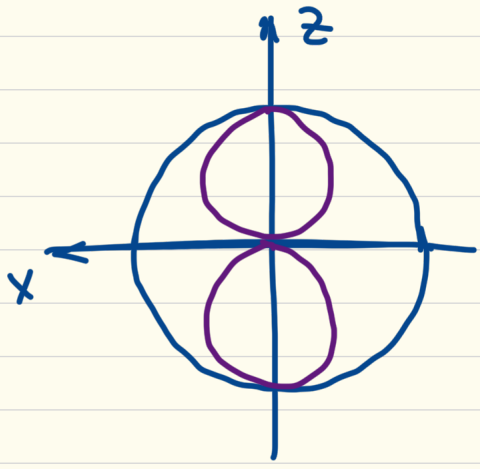
4.5. For Problem 4.1 determine the polarization of the radiated far-zone electric fields (E_θ, E_ϕ) and normalized amplitude pattern in the following planes:

(a) $\phi = 0°$ (b) $\phi = 90°$ (c) $\theta = 90°$

解.(a) $\phi=0°$时，$E_\psi = j\eta\frac{kIl_0 e^{-jkr}}{4\pi r}\sqrt{1-\sin^2\theta}$

$= j\eta\frac{kIl_0 e^{-jkr}}{4\pi r}\cos\theta$

$E_\theta = 0$



Normalized amplitude pattern:

4.31. A base-station cellular communication system utilizes arrays of $\lambda/2$ dipoles as transmitting and receiving antennas. Assuming that each element is *lossless* and that the *input power* to each of the $\lambda/2$ dipoles is 1 watt, determine at *1,900 MHz* and a distance of *5 km* the maximum

(a) radiation intensity. *Specify also the units.*

(b) radiation density (*in watts/m*^{2})

for each $\lambda/2$ *dipole*. This determines the safe level for human exposure to EM radiation.

解: (a). lossless => $P_{rad} = P_{in} = 1\,W$ $D_0 = 1.643$

$$U_{avg} = \frac{P_{rad}}{4\pi} = 0.07958\ w/sr$$

$$\Rightarrow U_{max} = D_0 U_{avg} = D_0 \frac{P_{rad}}{4\pi} = 0.1307\ W/sr$$

(b) $$W_{max} = \frac{U_{max}}{r^2} = 5.230 \times 10^{-9}\ w/m^2$$

$$W_{avg} = \frac{U_{avg}}{r^2} = 3.183 \times 10^{-9}\ w/m^2$$